

Extended Kalman Filtering for State Estimation in Fed-Batch Bioreactor Systems Under Varying Noise Regimes

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Abstract

This project applies Kalman filtering techniques to the problem of state estimation in a fed-batch bioreactor process model. A simplified system of ordinary differential equations governs biomass growth, substrate consumption, and product formation under Monod kinetics. Because only a subset of bioreactor states are directly measurable, and all measurements carry noise, the Kalman filter provides a principled framework for fusing heterogeneous sensor data to recover the true process trajectory. Online measurements from capacitance (viable cell concentration proxy) and Raman spectroscopy (glucose proxy) are combined with sparse offline assays to form a mixed-frequency estimation problem. The project investigates how process noise intensity, measurement noise covariance structure, and sensor availability influence estimator performance. Both the linear Kalman filter (applied to a locally linearized system) and the extended Kalman filter (EKF) for the full nonlinear dynamics are implemented and compared. This project connects recursive state estimation to the broader theory of stochastic processes, with particular emphasis on the martingale structure of the innovations sequence and its role as a model diagnostic.

1 Project Description

State estimation is a fundamental problem in stochastic process theory: given noisy, partial observations of a dynamical system, what is the best estimate of the system’s true state? The Kalman filter provides the minimum-variance unbiased estimate for linear–Gaussian systems and, through extensions such as the EKF, can be applied to nonlinear dynamics common in engineering practice.

Fed-batch bioreactors present a compelling application because:

- The underlying dynamics are nonlinear (Monod growth kinetics, time-varying feed inputs).
- Only a subset of states are measured online (e.g., capacitance as a proxy for viable cell concentration, Raman spectroscopy as a proxy for glucose), while others require infrequent offline assays (titer, metabolite concentrations).
- Measurements arrive at mixed frequencies and carry distinct noise characteristics (sensor drift, assay variability, sampling error).

The central question of this project is:

How do the structure and intensity of process and measurement noise affect the ability of the Kalman filter to recover the true bioreactor state trajectory, and how does sensor configuration influence observability and estimation accuracy?

This connects directly to course themes: the Kalman filter as a minimum-variance recursive estimator, the innovations sequence as a martingale, and the interplay between signal and noise in stochastic systems.

2 Mathematical Formulation

2.1 Continuous-Time Process Model

The bioreactor state vector is $\mathbf{x}(t) = [X_v(t), S(t), P(t)]^\top$, where X_v is viable cell concentration, S is limiting substrate (glucose) concentration, and P is product (e.g., monoclonal antibody titer). The process dynamics are described by the following equations:

$$\frac{dX_v}{dt} = \mu(S) X_v - k_d X_v \quad (1)$$

$$\frac{dS}{dt} = -\frac{\mu(S) X_v}{Y_{X/S}} - m_S X_v + \frac{F_S(t)}{V(t)} \quad (2)$$

$$\frac{dP}{dt} = q_P X_v \quad (3)$$

where the specific growth rate follows Monod kinetics,

$$\mu(S) = \mu_{\max} \frac{S}{K_S + S}, \quad (4)$$

k_d is the cell death rate, $Y_{X/S}$ is the biomass-on-substrate yield, m_S is the maintenance coefficient, q_P is the specific productivity, $F_S(t)$ is the substrate feed rate (a known, time-varying input), and $V(t)$ is the culture volume.

2.2 Discrete-Time State-Space Representation

For Kalman filter implementation, the continuous dynamics are discretized via an Euler or Runge–Kutta scheme at sampling interval Δt :

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k) \quad (5)$$

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k) \quad (6)$$

where $f(\cdot)$ is the discretized nonlinear state transition, $\mathbf{u}_k$ is the known input vector (feed rate, temperature), $h(\cdot)$ is the measurement function mapping states to observables, $\mathbf{w}_k$ is process noise capturing model uncertainty, and $\mathbf{v}_k$ is measurement noise. The covariance matrices $\mathbf{Q}_k$ (process) and $\mathbf{R}_k$ (measurement) are the key parameters whose influence this project investigates.

2.3 Measurement Model

The available measurements and their relationship to the state vector are:

$$\mathbf{z}_k = \begin{bmatrix} z_k^{\text{cap}} \\ z_k^{\text{raman}} \\ z_k^{\text{VCC}} \\ z_k^{\text{gluc}} \\ z_k^{\text{titer}} \end{bmatrix} = \begin{bmatrix} \alpha X_v(t_k) \\ \beta S(t_k) \\ X_v(t_k) \\ S(t_k) \\ P(t_k) \end{bmatrix} + \mathbf{v}_k \quad (7)$$

- z^{cap} is a continuous online capacitance measurement (linear proxy for X_v with proportionality constant α and moderate noise)
- z^{raman} is a continuous online Raman spectroscopy measurement (linear proxy for glucose S with proportionality constant β and moderate noise)
- z^{VCC} is an offline cell count (available at sparse intervals with low noise) and z^{gluc} , z^{titer} are offline assay measurements (available at the same sparse intervals with low noise)

This multiple measurement structure, mixing two online sensors sampled at high frequency with sparse offline assays, is a natural setting for Kalman filter-based data fusion. The two online probes provide continuous but noisier readings of X_v and S , while the offline assays periodically anchor the estimates with higher-precision measurements.

2.4 Martingale Structure of the Innovations Sequence

The *innovations sequence* is defined as the difference between the observed measurement and the filter's predicted measurement:

$$\boldsymbol{\nu}_k = \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}). \quad (8)$$

When the filter model is correctly specified (i.e., the true dynamics match the assumed model and the noise statistics are accurate), the innovations form a *martingale difference sequence*: $E[\boldsymbol{\nu}_k | \mathcal{F}_{k-1}] = \mathbf{0}$, where $\mathcal{F}_{k-1}$ is the filtration generated by all observations up to time $k-1$. So, the innovations are uncorrelated and zero-mean, with covariance

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k. \quad (9)$$

This martingale property is the central connection between Kalman filtering and stochastic process theory as covered in this course. It has two important consequences:

1. **Optimality criterion:** The Kalman filter is the unique linear filter that produces innovations with zero conditional mean, it extracts all information from the measurements, leaving only unpredictable noise.
2. **Model diagnostic:** If the innovations exhibit autocorrelation, non-zero mean, or variance inconsistent with $\mathbf{S}_k$, the martingale property is violated and the filter model is misspecified. This provides a practical diagnostic for detecting model mismatch, incorrect noise tuning, or sensor degradation.

The normalized innovations,

$$\bar{\boldsymbol{\nu}}_k = \mathbf{S}_k^{-1/2} \boldsymbol{\nu}_k, \quad (10)$$

should be approximately standard normal and uncorrelated under correct model specification. The project will use whiteness tests (autocorrelation function, Ljung-Box statistic) on the normalized innovations as a diagnostic tool across the noise regime experiments.

3 Extended Kalman Filter for Nonlinear Dynamics

Because the bioreactor model (1)–(3) is nonlinear, the standard Kalman filter cannot be applied directly. The extended Kalman filter linearizes the dynamics about the current state estimate at each time step. Let

$$\mathbf{F}_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k}}, \quad \mathbf{H}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k-1}} \quad (11)$$

denote the Jacobians of the state transition and measurement functions, respectively. For the bioreactor system, the state-transition Jacobian $\mathbf{F}_k$ involves partial derivatives of the Monod kinetics with respect to S , introducing coupling between the biomass and substrate estimation errors.

The EKF recursion then follows the standard predict–update cycle using these linearized matrices, with the nonlinear functions $f(\cdot)$ and $h(\cdot)$ used for the mean propagation.

4 Noise Regime Analysis

A central focus of this project is a systematic study of how noise structure affects estimation quality. The following noise scenarios will be investigated:

4.1 Process Noise Intensity

Varying $\mathbf{Q}$ from low to high simulates conditions ranging from a well-characterized process (where the ODE model closely matches reality) to a poorly known or highly variable process. The project will examine how increasing $\mathbf{Q}$ widens the estimator’s uncertainty bounds ($\mathbf{P}_{k|k}$) and shifts the balance between trusting the model versus trusting the measurements.

4.2 Measurement Noise Structure

Different diagonal entries of $\mathbf{R}$ reflect realistic differences in sensor quality:

- Online capacitance: high frequency, moderate noise (R^{cap} moderate-to-large).
- Online Raman spectroscopy: high frequency, moderate noise (R^{raman} moderate-to-large).
- Offline VCC, glucose, titer: low frequency but more precise (R^{offline} small).

The project will explore how the Kalman gain adapts to weight precise-but-sparse offline measurements against noisy-but-continuous online signals from capacitance and Raman probes.

4.3 Sensor Availability and Observability

By selectively removing measurements (e.g., running with only the two online probes and no offline assays, or with only one online probe), the project will examine the system’s observability. The observability Gramian,

$$\mathcal{O} = \sum_{k=0}^N (\mathbf{F}^k)^\top \mathbf{H}^\top \mathbf{H} \mathbf{F}^k, \quad (12)$$

determines whether the full state can be reconstructed from the available measurements. When $\mathcal{O}$ is rank-deficient, certain states become unobservable, and the corresponding entries of $\mathbf{P}_{k|k}$ will not converge.

5 Theoretical Focus

The theoretical component of the project will emphasize the following ideas from the course:

- **Martingale innovations:** The innovations sequence $\{\boldsymbol{\nu}_k\}$ as a martingale difference sequence, its connection to the optional sampling theorem, and its use as a model diagnostic via whiteness testing.

- **Minimum-variance estimation:** The Kalman filter as the optimal linear estimator minimizing the trace of the error covariance $\mathbf{P}_{k|k}$, and how this optimality degrades under model mismatch.
- **Optimality and the Riccati equation:** The steady-state error covariance satisfying the discrete algebraic Riccati equation (DARE), and conditions under which $\mathbf{P}_{k|k}$ converges.
- **Stochastic process connections:** Interpreting the process noise as a discretized diffusion perturbation and relating the continuous-time filtering problem to Itô calculus and the Kalman–Bucy filter.
- **Observability and controllability:** How sensor placement and process inputs determine what can and cannot be estimated.

6 Computational Component

All simulations will be implemented in Python using NumPy, SciPy, and Matplotlib. The computational workflow is:

1. **Ground truth generation:** Solve the ODE system (1)–(3) using `scipy.integrate.solve_ivp` (RK45) to produce the “true” bioreactor trajectory for a 14-day fed-batch process.
2. **Synthetic measurement generation:** Sample measurements from the true trajectory at appropriate frequencies, adding Gaussian noise according to the specified $\mathbf{R}$ matrix.
3. **EKF implementation:** Run the predict–update recursion on the synthetic measurements to produce state estimates $\hat{\mathbf{x}}_{k|k}$ and covariance histories $\mathbf{P}_{k|k}$.
4. **Performance evaluation:** Compute root-mean-square estimation error (RMSE) and compare the estimated $\pm 2\sigma$ confidence bounds against the true trajectory.
5. **Parametric studies:** Sweep across noise regimes ($\mathbf{Q}$, $\mathbf{R}$), sensor configurations, and model mismatch scenarios.

6.1 Pseudocode: Extended Kalman Filter

Algorithm 1 Extended Kalman Filter for Bioreactor State Estimation

Require: Initial state estimate $\hat{\mathbf{x}}_{0|0}$, initial covariance $\mathbf{P}_{0|0}$, process noise $\mathbf{Q}$, measurement noise $\mathbf{R}$

Ensure: State estimates $\{\hat{\mathbf{x}}_{k|k}\}_{k=1}^N$, covariance history $\{\mathbf{P}_{k|k}\}_{k=1}^N$

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1: for  $k = 1, 2, \dots, N$  do
2:   — Predict Step —
3:   Propagate state through nonlinear model:
4:      $\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{u}_{k-1})$  (RK4 integration of ODEs)
5:   Compute state-transition Jacobian:
6:      $\mathbf{F}_{k-1} = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k-1|k-1}}$ 
7:   Propagate covariance:
8:      $\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k-1}^\top + \mathbf{Q}$ 
9:   — Update Step —
10:  if measurement  $\mathbf{z}_k$  is available at time  $t_k$  then
11:    Compute measurement Jacobian:
12:       $\mathbf{H}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k-1}}$ 
13:    Compute innovations:
14:       $\boldsymbol{\nu}_k = \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})$ 
15:    Compute innovations covariance:
16:       $\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}$ 
17:    Compute Kalman gain:
18:       $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1}$ 
19:    Update state estimate:
20:       $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \boldsymbol{\nu}_k$ 
21:    Update covariance (Joseph form for numerical stability):
22:       $\mathbf{M} = \mathbf{I} - \mathbf{K}_k \mathbf{H}_k$ 
23:       $\mathbf{P}_{k|k} = \mathbf{M} \mathbf{P}_{k|k-1} \mathbf{M}^\top + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^\top$ 
24:  else
25:     $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1}, \quad \mathbf{P}_{k|k} = \mathbf{P}_{k|k-1}$ 
26:  end if
27: end for
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6.2 Pseudocode: Jacobian Computation for Bioreactor Model

Algorithm 2 Analytical Jacobian $\mathbf{F}_k$ for the Bioreactor ODE System

Require: Current state estimate $\hat{\mathbf{x}} = [X_v, S, P]^\top$, parameters $\mu_{\max}, K_S, k_d, Y_{X/S}, m_S, q_P$

Ensure: Jacobian matrix $\mathbf{F} \in \mathbb{R}^{3 \times 3}$

- 1: Compute Monod terms:
- 2: $\mu = \mu_{\max} S / (K_S + S)$
- 3: $\frac{\partial \mu}{\partial S} = \mu_{\max} K_S / (K_S + S)^2$
- 4: Assemble Jacobian of the continuous-time system:

$$\mathbf{J} = \begin{bmatrix} \mu - k_d & \frac{\partial \mu}{\partial S} X_v & 0 \\ -\frac{\mu}{Y_{X/S}} - m_S & -\frac{1}{Y_{X/S}} \frac{\partial \mu}{\partial S} X_v & 0 \\ q_P & 0 & 0 \end{bmatrix}$$

- 5: Discretize: $\mathbf{F} = \mathbf{I} + \Delta t \mathbf{J}$ *(first-order Euler approximation)*
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